

SHRUTHI KODATI

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Professional Summary

Software Engineer with hands-on experience building and delivering scalable applications from concept to production. Proven ability to design secure, maintainable systems, optimize application performance, and create intuitive user experiences. Experienced in collaborating with cross-functional teams to deliver reliable, high-quality software.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, C#, Shell

Frontend & Backend: React.js, Next.js, Redux, HTML5, CSS3, Tailwind CSS, Node.js, Express.js, Flask, RESTful API Design, GraphQL, WebSockets, Socket.io, MVC Architecture, Authentication & Authorization, JWT, OAuth 2.0

Databases: PostgreSQL, MySQL, Microsoft SQL Server, MongoDB, Redis, Schema Design, Data Modeling, Query Optimization, Indexing, Transactions

Cloud & DevOps: Docker, Git, GitHub, CI/CD Pipelines, AWS (EC2, S3), Microsoft Azure, Render, Vercel, Linux, Nginx, Postman

Software Engineering: Agile, SDLC, Code Reviews, Unit Testing

CS Fundamentals: Data Structures, Algorithms, Object-Oriented Programming (OOP), System Design, Distributed Systems, Concurrency, Time & Space Complexity

Security: Signal Protocol (X3DH + Double Ratchet), End-to-End Encryption, Ed25519, Zero-Knowledge Architecture, Replay Attack Prevention

Data & ML: Pandas, NumPy, Scikit-learn, PyTorch, ETL Pipelines, FAISS, BM25, Stream Processing, Geospatial Analytics

Professional Experience

Software Engineer, Graduate Assistant | *University of Illinois Chicago, College of Nursing*

Dec 2024 – May 2026

- Designed, developed, and deployed a production web application that automated clinical dietary workflows for 3+ research teams, reducing manual processing effort by 40% across 500+ dietary records per workflow cycle using Python, Flask, JavaScript, SQL, and RESTful APIs.
- Optimized 12+ RESTful API endpoints by improving SQL queries, relational schema design, and backend data processing, reducing average response times by 35% while supporting concurrent production workloads.
- Built responsive, accessible frontend interfaces with dynamic rendering, client-side validation, and real-time feedback across 8+ workflow views, enabling non-technical users to efficiently manage 10K+ records.
- Owned end-to-end application delivery by implementing secure authentication, CI/CD pipelines, cloud deployment, performance profiling, and cross-functional collaboration, delivering reliable software for 15+ active users.
- Partnered with nursing researchers across 10+ requirement sessions to translate clinical workflows into production software, delivering iterative enhancements through Agile development and version-controlled releases.

Software Engineer, Intern | *CVR College of Engineering, R&D Cell*

Jan 2023 – June 2024

- Developed a full-stack web application using React.js, Node.js, Express.js, MongoDB, and Tailwind CSS to digitize research administration workflows across 9 academic departments.
- Built reusable frontend components, RESTful APIs, and database-driven workflows supporting faculty profiles, research projects, publications, and document management for 100+ users.
- Collaborated with faculty and stakeholders to translate business requirements into scalable software solutions, improving application usability through responsive design, efficient database integration, and end-to-end testing.
- Implemented end-to-end software development workflows, including data validation, error handling, testing, and deployment, improving application reliability and maintainability.

Projects

Cipher-Signal Protocol Encrypted Messenger | *Node.js, Socket.io, Signal Protocol, Ed25519*

- Engineered a real-time end-to-end encrypted messaging platform by implementing the Signal Protocol (X3DH + Double Ratchet), supporting secure messaging, 25MB encrypted file transfers across 12 file types, and forward secrecy.
- Strengthened application security through Ed25519 signature verification, replay attack prevention, rate limiting, and persistent offline message queuing with ACK-based deduplication, ensuring exactly-once message delivery.

Multi-Modal RAG for SEC 10-K Financial Document Analysis | *Python, Flask, FAISS, BM25, LLMs*

- Architected a multi-modal RAG pipeline processing 150 SEC 10-K filings (8GB, 45K text chunks, 12K tables) using hybrid FAISS + BM25 retrieval, achieving 85% Table Recall@5 and a 12-point Exact Match improvement over text-only baselines.
- Developed intelligent query routing and citation-grounded REST APIs, while achieving 96% faithfulness with a 4.2-second median response time across FinQA, DocFinQA, and FinDER benchmarks.

SF Property Tax Analysis & Gentrification Risk Assessment | *Python, GeoPandas, REST APIs, Altair*

- Built a scalable data processing pipeline for 3.7M property records (6GB GeoJSON), optimizing ETL workflows with streaming parsing and memory-efficient transformations that reduced memory usage by 87% (5.7GB to 733MB).
- Implemented geospatial data processing and risk modeling services to analyze 93 San Francisco neighborhoods, generating interactive insights that uncovered 59.6% median property appreciation and supported evidence-based urban planning.

Education

Master of Science in Computer Science | *University of Illinois Chicago* | **GPA: 3.6/4.0**

Aug 2024 – May 2026

Bachelor of Technology in Computer Science | *CVR College of Engineering* | **GPA: 8.08/10**

Aug 2020 – May 2024